VANISHING OF THE CUP PRODUCT WITH THE CLASS OF A SUBSCHEME FROM VANISHING ON THE SUBSCHEME

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file weil.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the reduction to the blowing up through the injectivity axiom, the rewriting of the pulled back class, and the application of the divisor axiom.

Editorial note. The label lemma-A5-A6-imply names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source weil.tex –label lemma-A5-A6-imply –start 4825 –end 4850 –local-id 28168 returned RESULT=OK. Pre-assembly tag self-check: canonical tag resolved and not present in reference/public-items.jsonl.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Compatibility of the cycle class map with pullback along a blowing up. Same chapter, the lemma identifying the pullback of the class of a closed subscheme along a blowing up with the product of the class of the exceptional divisor and a cycle. Rewrites the pulled back class of the subscheme as a product of the class of the exceptional divisor with another class.
- The axiom on the cup product with the first Chern class of a divisor. Axiom (A4) on the vanishing of the cup product with the class of an effective Cartier divisor after restriction vanishes. Concludes the vanishing from the vanishing of the restriction to the exceptional divisor.
- Injectivity after pullback along a blowing up. Axiom (A7) on the injectivity of the pullback along a blowing up. Lifts the identity to be proven to the blowing up, where the computation is available.

Source reference. Original reference 0FIH: sources/stacks-weil/pinned-original-source.tex, source lines 4761–4761. Prerequisite pointer invoked at proof line(s) 4842; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Assume given data (D0), (D1), and (D2') satisfying axioms (A1) – (A4) and (A7). Let X be a smooth projective scheme over k. Let $Z \subset X$ be a smooth closed subscheme such that every irreducible component of Z has codimension r in*

Date: stacks-project:weil.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

X. Assume the class of $\mathcal{C}_{Z/X}$ in $K_0(Z)$ is the restriction of an element of $K_0(X)$. If $a \in H^*(X)$ and $a|_Z = 0$ in $H^*(Z)$, then $\gamma([Z]) \cup a = 0$.

Proof. Let $b : X' \to X$ be the blowing up. By (A7) it suffices to show that

$$b^*(\gamma([Z]) \cup a) = b^*\gamma([Z]) \cup b^*a = 0$$

By Lemma 0FIH we have

$$b^*\gamma([Z]) = \gamma(b^*[Z]) = \gamma([E] \cdot \theta) = \gamma([E]) \cup \gamma(\theta)$$

Hence because b^*a restricts to zero on E and since $\gamma([E]) = c_1^H(\mathcal{O}_{X'}(E))$ we get what we want from (A4). $\square$